

PHP + MySQL



Apache, PHP, MySQL

Download:

- Apache Web Server: <httpd.apache.org/download.cgi>
- PHP application server: www.php.net/downloads.php
- MySQL DBMS: www.mysql.com/downloads/
- LAMP: integration of Linux, Apache, MySQL, and PHP
- Packages (Apache + PHP + MySQL in a single installer):
 - WAMP (Windows), www.wampserver.com/en/
 - MAMP (Windows, Mac), www.mamp.info/en/
 - XAMPP (Windows, Mac, Linux), www.apachefriends.org
XAMPP installs *MariaDB* (mariadb.org), which is a fully compatible fork of MySQL led by its original developers

PHP

Variables:

```
$num = 58;  
$txt = "Hello World";
```

Operators:

practically, the same as JavaScript (and Java, C, and C++)

Conditional constructs

```
if (condition) {  
    ...;  
} elseif (condition) {  
    ...;  
} else {  
    ...;  
}
```

PHP

Arrays:

- `$cars = array("Saab", "Volvo", "BMW", "Toyota");`
- `$cars[0] = "Saab";`
`$cars[1] = "Volvo";`
`$cars[2] = "BMW";`
`$cars[3] = "Toyota";`

Associative arrays:

- `$ages = array("Peter"=>32, "Quagmire"=>30, "Joe"=>34);`
- `$ages['Peter'] = "32";`
`$ages['Quagmire'] = "30";`
`$ages['Joe'] = "34";`

Also multidimensional...

PHP

Loops:

```
while (condition) {  
    ...;  
}
```

```
do {  
    ...;  
}  
while (condition);
```

```
for (init; condition; increment) {  
    ...;  
}
```

```
foreach ($array as $value) {  
    ...;  
}
```

PHP

Functions:

```
<html>
  <body>
    <?php
      function add($x,$y) {
        $total = $x + $y;
        return $total;
      }
      echo "1 + 16 = " . add(1,16);
    ?>
  </body>
</html>
```

PHP

include () function:

```
...  
<body>  
  <div id="header">Header page 1</div>  
  <div id="content">Content</div>  
  <div id="menu">  
    <?php include("menu.html");?>  
  </div>  
</body>  
...
```

File menu.html

```
<h2>Menu</h2>  
<ul>  
  <li><a href="...">Link1</a></li>  
  <li><a href="...">Link2</a></li>  
  <li><a href="...">Link3</a></li>  
  <li><a href="...">Link4</a></li>  
  <li><a href="...">Link5</a></li>  
</ul>
```

File management:

- `$file = fopen("welcome.txt","r");`

(*r* = Read only, starts at the beginning of the file; *r+* = Read/Write, starts at the beginning of the file; *w* = write only, opens and clears the contents of file or creates a new file if it doesn't exist; *w+* = read/write, opens and clears the contents of file or creates a new file if it doesn't exist; *a* = Append, opens and writes to the end of the file or creates a new file if it doesn't exist; *a+* = Read/Append, preserves file content by writing to the end of the file; *x* = Write only, creates a new file, returns FALSE and an error if file already exists; *x+* = Read/Write, creates a new file, returns FALSE and an error if file already exists)

- `fclose($file);`

- Row-by-row reading:

```
while(!feof($file)) {  
    echo fgets($file). "<br>";  
}
```


Opening a connection:

```
<?php
$con = mysqli_connect("localhost","peter","abc123");
if (!$con) {
    die('Could not connect: ' . mysqli_error($con));
}
?>
```

Closing a connection:

```
mysqli_close($con);
```

Database creation:

```
if (mysqli_query($con,"CREATE DATABASE my_db")) {
    echo "Database created";
} else {
    echo "Error creating database: " . mysqli_error($con);
}
```

Note: `mysql_connect`, `mysql_close`, `mysql_select_db`, `mysql_query`, `mysql_fetch_array`, etc., are deprecated. The *MySQLi* extension is designed to work with MySQL version 4.1.13 or newer

Table creation:

```
CREATE TABLE table_name
(
    column_name1 data_type,
    column_name2 data_type,
    column_name3 data_type,
    ....
)
```

Entering data:

```
INSERT INTO table_name (column1, column2, column3,...)
VALUES (value1, value2, value3,...)
```

Example (table creation):

```
<?php
...
mysqli_select_db($con,"my_db");
$sql = "CREATE TABLE People
(
    FirstName varchar(15),
    LastName varchar(15),
    Age int
) ";

// Execute query
mysqli_query($con,$sql);

mysqli_close($con);
?>
```

Example (data entering):

```
<?php
    $con = mysqli_connect("localhost","peter","abc123");
    if (!$con) {
        die('Could not connect: ' . mysqli_error($con));
    }
    mysqli_select_db($con,"my_db");

    mysqli_query($con,"INSERT INTO People (
        FirstName, LastName, Age)
        VALUES ('Peter', 'Griffin', '35')");

    mysqli_query($con,"INSERT INTO People (FirstName, LastName,
        Age) VALUES ('Glenn', 'Quagmire', '33')");

    mysqli_close($con);
?>
```

Example (entering data received from a form):

```
<form action="insert.php" method="post">
  Firstname: <input type="text" name="firstname">
  Lastname: <input type="text" name="lastname">
  Age: <input type="text" name="age">
  <input type="submit">
</form>
<?php
  $con = mysqli_connect("localhost","peter","abc123");
  if (!$con) { die('Could not connect: ' . mysqli_error($con)); }
  mysqli_select_db($con,"my_db");
  $sql="INSERT INTO People (FirstName, LastName, Age)
        VALUES ('$_POST[firstname]','$_POST[lastname]',
                  '$_POST[age]')";
  if (!mysqli_query($con,$sql))
    { die('Error: ' . mysqli_error($con)); }
  mysqli_close($con)
?>
```

MySQL

Example (select):

```
<?php
    $con = mysqli_connect("localhost","peter","abc123");
    if (!$con) {
        die('Could not connect: ' . mysqli_error($con));
    }

    mysqli_select_db($con,"my_db");

    $result = mysqli_query($con,"SELECT * FROM People");

    while($row = mysqli_fetch_array($result,MYSQLI_ASSOC)) {
        echo $row['FirstName'] . " " . $row['LastName'];
        echo "<br>";
    }
    mysqli_close($con);
?>
```